

# AIRBUS

**Airbus Amber**

**AIMS05-02-004**

Issue 3

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## **Airbus Material Specification**

**Glass fibre reinforced epoxy prepreg**

**120 °C curing class/4 harness satin weave (105 g/m<sup>2</sup>)**

**Non-structural application**

### **Export Control**

|                                                                                                                                                                           |
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## 1 Scope

This Material Specification defines the requirements of a group of materials consisting of epoxy resin reinforced with 105 g/m<sup>2</sup> glass fibre fabric, 120 °C nominal cure temperature, qualified according to AIMS05-02-000 Part 1 and Part 2 for aerospace application.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

## 2 Application

This material is used for the manufacture of non-structural components.

## 3 Normative references

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

|                      |                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ABS5009              | Product Standard - EP-Prepreg Glass fibre fabric preimpregnated with epoxy resin.                                                                         |
| AIMS05-02-000 Part 1 | Technical Specification - Glass Fibre Reinforced Epoxy Prepreg; Part1: General requirements.                                                              |
| AIMS05-02-000 Part 2 | Technical Specification - Glass Fibre Reinforced Epoxy Prepreg; Part 2: Qualification program / Batch release testing. Fabric/120 °C - Curing class.      |
| AITM1-0001           | Test Method – Fibre reinforced plastics – Determination of mechanical degradation due to chemical paint strippers.                                        |
| AITM1-0003           | Test Method – Determination of the glass transition temperatures.                                                                                         |
| AITM3-0001           | Test Method – Analysis of thermoset systems by High performance liquid chromatographie.                                                                   |
| AITM3-0002           | Test Method – Analysis of non-metallic materials (uncured) by Differential scanning calorimetry.                                                          |
| AITM3-0003           | Test Method- Analysis of organic compounds by infrared spectroscopy                                                                                       |
| AITM3-0004           | Test Method - Determination of gel time and viscosity.                                                                                                    |
| AITM3-0008           | Test Method – Determination of the extent of cure by Differential scanning calorimetry                                                                    |
| EN2329               | Aerospace series - Textile glass fibre preimpregnates - Test method for the determination of mass per unit area.                                          |
| EN2330               | Aerospace series – Textile glass fibre preimpregnates - Test method for the determination of the content of volatile matter.                              |
| EN2331               | Aerospace series – Textile glass fibre preimpregnates – Test method for the determination of the resin and fibre content and mass of fibre per unit area. |
| EN2332               | Aerospace series – Textile glass fibre preimpregnates – Test method for the determination of the resin flow.                                              |
| EN2377               | Aerospace series – Glass fibre reinforced plastics – Test method –Determination of apparent Interlaminar shear strength.                                  |
| ISO1183              | Plastics – Methods for determining the density of non-cellular plastics.                                                                                  |
| ISO1889              | Reinforcement yarns - Determination of linear density.                                                                                                    |

## 4 Requirements

### 4.1 General

Specification values are minimum except where stated.

### 4.2 Specific

#### 4.2.1 Material description

The material description is defined in Table 1

**Table 1: Material description**

| Characteristic                   | Requirement                                                                            |
|----------------------------------|----------------------------------------------------------------------------------------|
| Technical Specification          | AIMS05-02-000 Part 1 / AIMS05-02-000 Part 2                                            |
| Material type                    | Glass fibre fabric reinforced epoxy prepreg                                            |
| Fibre                            | Glass fibre, type E                                                                    |
| Delivery form                    | Preimpregnated fabric                                                                  |
| Fabric                           | Style 120 4 Harness Satin Weave, 105 g/m <sup>2</sup><br>AMS-C-9084C Type III, Class 2 |
| Finish                           | Epoxy compatible                                                                       |
| Matrix                           | Modified epoxy resin                                                                   |
| Resin content by weight, uncured | 42 % nominal                                                                           |

#### 4.2.2 Processing information

The cure cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification programme and shall be used to produce all test specimens for qualification testing.

#### 4.2.3 Release, screening and qualification requirements

The testing requirements for release, screening and material qualification are specified in AIMS05-02-000 Part 1 and Part 2.

The acceptance values are specified in Table 3 and Table 4 of this specification.

#### 4.2.4 Work life and storage life

See Table 2

**Table 2: Worklife and storage life**

| No | Properties                                   | Unit | Requirements |
|----|----------------------------------------------|------|--------------|
| 1  | Work life at 30 °C / 75 % R.H.               | Days | ≥10          |
| 2  | Storage life at -18 °C from date of shipment | Days | ≥12          |

## 5 Designation

See relevant ABS5009

## 6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-02-000.

**Table 3: Physical and physical-chemical properties**

| Test no                                                                                                                                                                          | Property                                      | Test method                              | Unit              | Requirements <sup>a)</sup> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------|-------------------|----------------------------|
| Uncured properties                                                                                                                                                               |                                               |                                          |                   |                            |
| 1                                                                                                                                                                                | Prepreg areal weight                          | EN2329                                   | g/m <sup>2</sup>  | 190 ± 15                   |
| 2                                                                                                                                                                                | Fibre areal weight                            | EN2331                                   | g/m <sup>2</sup>  | 105 ± 6                    |
| 3                                                                                                                                                                                | Resin density                                 | ISO1183 A                                | g/cm <sup>3</sup> | b)                         |
| 4                                                                                                                                                                                | Fibre density                                 | ISO1889                                  | g/cm <sup>3</sup> | b)                         |
| 5                                                                                                                                                                                | Volatile content                              | EN2330                                   | %                 | 2,0 maximum                |
| 6                                                                                                                                                                                | Resin content                                 | EN2331                                   | %                 | 42 ± 3                     |
| 7                                                                                                                                                                                | Infrared Spectroscopy                         | AITM3-0003                               | -                 |                            |
| 8                                                                                                                                                                                | High Performance Liquid Chromatography (HPLC) | AITM3-0001                               | -                 | b)                         |
| 9                                                                                                                                                                                | Differential Scanning Calorimetry (DSC)       | AITM3-0002<br>(10 °C/min)                | -                 | b)                         |
| 10                                                                                                                                                                               | Viscosity                                     | AITM3-0004<br>Method A                   | -                 | b)                         |
| 11                                                                                                                                                                               | Gel time                                      | AITM3-0004<br>(Method B or C)            | Min               | b)                         |
| 12                                                                                                                                                                               | Resin flow                                    | EN2332                                   | %                 | b)                         |
| 13                                                                                                                                                                               | Tack                                          | As defined in<br>AIMS05-02-000<br>Part 1 | -                 | Pass                       |
| Cured properties                                                                                                                                                                 |                                               |                                          |                   |                            |
| 14                                                                                                                                                                               | Cured ply thickness                           | AIMS05-02-000<br>Part 1                  | mm                | b)                         |
| 15                                                                                                                                                                               | Degree of cure                                | AITM3-0008                               | %                 | ≥ 90                       |
| <sup>a)</sup> The requirements are to be applied to the average values.<br><sup>b)</sup> To be defined during qualification and recorded in the Qualification Test Report or IPS |                                               |                                          |                   |                            |

**Table 4: Requirements for cured prepreg - Mechanical properties**

| Test no                                                                                                                        | Properties                                |                                       | Test method                         | Units | Test condition             | Test temperature (°C) | Requirements <sup>a)</sup> |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------|-------------------------------------|-------|----------------------------|-----------------------|----------------------------|
| 1                                                                                                                              | Glass transition temperature (Tg) (Onset) |                                       | AITM 1-0003                         | °C    | Dry <sup>b)</sup>          | -                     | ≥ 120                      |
|                                                                                                                                |                                           |                                       |                                     |       | 70 °C / 85 %RH equilibrium | -                     | ≥ 100                      |
| 2                                                                                                                              | Interlaminar shear strength (ILSS) (warp) |                                       | EN2377                              | MPa   | Ambient                    | -55                   | 60                         |
|                                                                                                                                |                                           |                                       |                                     |       |                            | RT                    | 42                         |
|                                                                                                                                |                                           |                                       |                                     |       |                            | 80                    | 25                         |
|                                                                                                                                |                                           |                                       |                                     |       | 70 °C / 85 %RH equilibrium | 80                    | 15                         |
| 3                                                                                                                              | Resistance to test fluids [ILSS (warp)]   | Water<br>50 °C/1000 hours             | EN2377 and AIMS 05-02-000 Part 1    | MPa   | -                          | 80                    | c) d)                      |
| 4                                                                                                                              |                                           | Skydrol<br>70 °C/1000 hours           |                                     | MPa   | -                          | 80                    | c) d)                      |
| 5                                                                                                                              |                                           | Skydrol / water<br>50 °C / 1000 hours |                                     | MPa   | -                          | 80                    | c) d)                      |
| 6                                                                                                                              |                                           | Fuel<br>RT/1000 hours                 |                                     | MPa   | -                          | 80                    | c) d)                      |
| 7                                                                                                                              |                                           | Solvent<br>RT/1 hour                  |                                     | MPa   | -                          | RT                    | c) d)                      |
| 8                                                                                                                              | Paint stripper (phenolic)                 |                                       | AITM 1-0001 + AIMS 05-02-000 Part 1 | MPa   | -                          | RT                    | d)                         |
| 9                                                                                                                              | Paint stripper (non-phenolic)             |                                       |                                     | MPa   | -                          | RT                    | d)                         |
| a) The requirements are to be applied to the minimum individual values.                                                        |                                           |                                       |                                     |       |                            |                       |                            |
| b) Drying conditions: to be agreed between the Qualification Project Leader and the Manufacturer.                              |                                           |                                       |                                     |       |                            |                       |                            |
| c) Values after conditioning will have no significant decrease compared to the reference value tested at the same temperature. |                                           |                                       |                                     |       |                            |                       |                            |
| d) Results to be reported during qualification.                                                                                |                                           |                                       |                                     |       |                            |                       |                            |



Record of Revisions

| Issue        | Clause modified                                    | Description of Modification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1<br>06.2001 | -                                                  | New Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2<br>03.2019 | Title<br>2<br>3<br>4.2.1<br>Table 3<br><br>Table 4 | Updated to latest template and reformatted as required. Title updated<br>Application updated<br>Normative references updated<br>Table 1: Resin content updated to (39 - 42) %<br>Prepreg area mass updated to 190 ± 15 g/m2<br>Fiber area mass updated to 105 ± 6 g/m2<br>Resin content updated to ((39 - 42) ± 3) %<br>Viscosity and degree of cure added<br>Moisture absorption and fluid tightness tests deleted<br>Notes updated<br>Tensile strength and modulus requirements deleted<br>Glass transition temperature added<br>ILSS requirement at (70°C/85%RH, RT) deleted<br>Resistance to test fluids (Interlaminar shear (warp)) test requirements updated<br>Notes updated |
| 3<br>03.2025 | All<br>Tables<br>4.2.1<br>6                        | ECCN details updated.<br>Foot note numbers changed as letters.<br>Table 1: Resin content by weight, Uncured was changed to 42 % nominal<br>Table 3: Resin content is changed to only 42 ± 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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